

EDUCATION

Ph.D. in Electrical and Computer Engineering Worcester Polytechnic Institute, Worcester, MA Advisor: Prof. Berk Sunar	Aug 2019 - Dec 2024
M.S. in Electrical and Computer Engineering Worcester Polytechnic Institute, Worcester, MA	Aug 2019 - May 2022
B.S. in Electrical and Electronics Engineering Middle East Technical University, Ankara, TR	Sep 2014 - Jun 2019

EXPERIENCE

Research Associate Worcester Polytechnic Institute	Worcester, MA Feb 2025 - Current
– Open Tools, Interfaces and Metrics for Implementation Security Testing – optimist-ose.org	
Hardware Security Intern NVIDIA	Santa Clara, CA 2nd - May 2024 - Aug 2024 1st - May 2022 - Aug 2022
– Evaluated EM side-channel attacks on GPUs for ML model extraction and defenses against them. – Developed a ML-based tool for automatic detection of side-channel leakage on CUDA libraries using GPU hardware performance counters.	
Research and Teaching Assistant Vernam Lab, Worcester Polytechnic Institute	Worcester, MA Aug 2019 - Dec 2024
– Research on Microarchitectural side-channel, fault injection attacks, and the applications of Transformer models, generative AI, and reinforcement learning on discovery, detection, and mitigation of these attacks.	
Research Intern Vernam Lab, Worcester Polytechnic Institute	Worcester, MA June 2018 - Sep 2018
– Co-authored a paper on ML-based website fingerprinting with cache timing side-channel on Google Nexus 5X.	
Reliability Engineering Intern Voltage Performance and Standby Lab, Arçelik A.Ş.	Istanbul, TR Jul 2017 - Sep 2017

SKILLS

HW Security	Microarchitectural side-channels, Transient Execution Attacks, Rowhammer, Voltage glitching, EM side-channels, TVLA, Oscilloscope
Programming:	ARM/x86 Assembly, C/C++, Python, CUDA, Bash, MATLAB, Git
Binary Analysis:	Linux Perf, PAPI, Intel Pin, objdump, Ghidra
Libraries:	PyTorch, TensorFlow, Keras, Scipy

RESEARCH

- M. C. Tol**, K. Derya, and B. Sunar, “Reinforcement Learning-based Discovery of Microarchitectural Vulnerabilities”, 2025 [*Under Review*].
- K. Derya, **M. C. Tol**, and B. Sunar, “Fault+Probe: A Generic Rowhammer-based Bit Recovery Attack”, 2024 [*Under Review*].
- A. J. Adiletta, **M. C. Tol**, and B. Sunar, “LeapFrog: The Rowhammer Instruction Skip Attack”, *hardwear.io USA, Santa Clara, CA, 2024., EuroSP Venice, Italy 2025*.
- M. C. Tol**, and B. Sunar, “ZeroLeak: Automated Side-Channel Patching in Source Code Using LLMs”, *Real World Crypto, Toronto, Canada, 2024. and ESORICS, Bydgoszcz, Poland, 2024*.
- A. J. Adiletta*, **M. C. Tol***, Y. Doroz, and B. Sunar, “Mayhem: Targeted Corruption of Register and Stack Variables”, in *Proceedings of the 2024 ACM Asia CCS, Singapore, 2024*. 3rd place in poster competition @NEHWS23.
- M. C. Tol**, S. Islam, A. J. Adiletta, B. Sunar, and Z. Zhang, “Don’t Knock! Rowhammer at the Backdoor of DNN Models”, in *Proceedings of the 2023 IEEE/IFIP International Conference on Dependable Systems and Network, Porto, Portugal, 2023*. 1st place in poster competition @NEHWS22.
- K. Mus, Y. Doroz, **M. C. Tol**, K. Rahman, and B. Sunar, “Jolt: Recovering TLS Signing Keys via Faults”, in *Proceedings of the 2023 IEEE Symposium on Security and Privacy, San Francisco, CA, 2023*.
- M. C. Tol**, B. Gulmezoglu, K. Yurtseven, and B. Sunar, “FastSpec: Scalable Generation and Detection of Spectre Gadgets Using Neural Embeddings”, in *Proceedings of the 2021 IEEE European Symposium on Security and Privacy, Vienna, Austria, 2021*.
- L. Amorós, S. M. Hafiz, K. Lee, and **M. C. Tol**, “Gimme That Modell: A Trusted ML Model Trading Protocol”, *Protecting Privacy through Homomorphic Encryption, 2021*.
- B. Gulmezoglu, A. Zankl, **M. C. Tol**, S. Islam, T. Eisenbarth, and B. Sunar, “Undermining User Privacy on Mobile Devices Using AI”, in *Proceedings of the 2019 ACM Asia CCS, Auckland, New Zealand, 2019*.

AWARDS

Best Poster Award <i>New England Hardware Security Day</i>	Apr 2022
IoT Application Award <i>The Senior Engineering Design Course Committee, METU EE</i>	Jun 2019
Honor Student <i>Middle East Technical University</i>	Jun 2019

SERVICES

Reviewer at IEEE Transactions on Computers	January 2025
Artifact Evaluation Committee Member at ACM CCS	May 2024
Reviewer at IEEE Transactions on Information Forensics and Security	December 2024
Reviewer at IEEE Transactions on Emerging Topics in Computing	October 2023
Reviewer at The Computer Journal, Oxford University Press	May 2023